

Shunpeng Bai

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Education

Huazhong Agricultural University

B.S. IN BIOENGINEERING

• GPA: 3.39/4.0

Wuhan, China

September 2021 - June 2026

Experience

Green Pesticide Chemistry Laboratory

HZAU, Wuhan, China

DIRECTED BY DR. HUAILONG TENG

Apr. 2022 - Oct. 2023

- “Synthesis of remote fluoroalkenyl ketones by photo-induced ring-opening addition of cyclic alkoxy radicals to fluorinated alkenes” D Du, H Peng, L He, S Bai, Z Li, H Teng - *Organic & Biomolecular Chemistry*, 2022
 - **Description:** Synthesized several compounds, and learned many experimental skills, including synthesis, isolation, characterization.

Biocatalysis Laboratory

HZAU, Wuhan, China

DIRECTED BY DR. SHUKE WU

Jun. 2022 - Oct. 2023

- Scientific research and training project was initiated titled “Chemico-enzyme cascade method for green and efficient preparation of Nepetalactone and its analogues”
 - **Description:** Directed evolution of P450-BM3 (CYP102A1) to obtain site selectivity for adipochain oxidation. I learned lots of technologies from DNA to protein, modifying protein, simulation *in silico* and detection of enzyme activity.

Natural Pharmaceutical Chemistry Laboratory

HZAU, Wuhan, China

DIRECTED BY ASSISTANT PROFESSOR SHENGZHEN XU

Oct. 2023 - June 2026

- “An organic small molecule fluorescent probe for nondestructive detection of Zn^{2+} in plants” J Zheng#, S Li#, S Bai#, X Liu, R Sun, S Xu, W Qu-*Tetrahedron Letters*.2025
 - **Description:** In this project, I undertook the work of theoretical calculation, including conformational search, DFT optimization and single point energy calculation for the probes before and after the combination. The reason why HOMO-LUMO gap becomes larger is explained. I learned a lot of theoretical calculation software and the use of computer cluster, and I wrote some scripts for convenient research.

Shen-Lab

NTU, Singapore

DIRECTED BY ASSISTANT PROFESSOR XIAOTAO SHEN

Mar. 2025 - June 2026

- “LagCI Enables Inference of Temporal Causal Relationships from Dense Multi-Omic Time Series” Y Ge#, S Bai#, Z Qiang#, Y Liu#, Y Wu, X Shen - *Briefings in Bioinformatics*, 2026
 - **Description:** As a co-first author, I co-designed the lagCI method, developed the R package and its Docker-based implementation, and performed case-study analyses. The preprint is on *bioRxiv*; the manuscript is currently under submission to a peer-reviewed journal.

Aitao Li Group

HUBU, Wuhan, China

DIRECTED BY PROFESSOR AITAO LI

Sep. 2025 - Jan. 2026

- “Asymmetric non-natural reactions with heme-containing enzymes”
 - **Description:** I performed substrate screening and directed evolution of the enzymes, and used computational simulation of the reaction mechanism to guide protein engineering.

Yajie Wang Group

Westlake University, Hangzhou, China

VISITING STUDENT, DIRECTED BY ASSISTANT PROFESSOR YAJIE WANG

Jan. 2026 - present

- Visiting student in the group of Assistant Professor Yajie Wang.

Co-first authors

Awards & Honours

- 2024 Gold Award in the Synthetic Biology Innovation Competition
- Organic lecture titled “Understanding organic reaction from Markov addition reaction mechanism” for undergraduate student
- Third Prize in the 15th “Huazhong Cup” College Students’ Mathematical Modeling competition
- Served as a peer tutor for the chemistry course at Huazhong Agricultural University (2022.2-2022.7)

Skills

Programming Python, R, Slurm, LaTeX, Bash, Git, Docker, GitHub Action
Spoken Languages Chinese (Native), English

Last Updated: 2026-09-11